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PUBLICATIONS

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PUBLICATIONS

- [95] Paul Zinn-Justin. q -difference equations for two classes of pattern avoiding permutations, 2026. [arXiv:2608.09078](#).
- [94] Geordie Williamson, Oded Yacobi, and Paul Zinn-Justin. Generating Hadamard matrices with transformers, 2026. [arXiv:2604.11101](#).
- [93] Ajeeth Gunna, Michael Wheeler, and Paul Zinn-Justin. Inhomogeneous q -Whittaker polynomials I: Duality and expansions, 2025. [arXiv:2512.04468](#).
- [92] Bruce W. Westbury and Paul Zinn-Justin. A uniform R -matrix for adjoint representations. *Bulg. J. Phys.*, 52(1):159–164, 2025.
- [91] Allen Knutson and Paul Zinn-Justin. Hybrid pipe dreams for the lower-upper scheme, 2025. [arXiv:2509.01857](#).
- [90] Ajeeth Gunna, Michael Wheeler, and Paul Zinn-Justin. Structure constants for spin Hall–Littlewood functions, 2025. [arXiv:2504.19205](#).
- [89] Allen Knutson and Paul Zinn-Justin. Generic pipe dreams, lower-upper varieties, and Schwartz–MacPherson classes. In *Proceedings of the 37th Conference on Formal Power Series and Algebraic Combinatorics (San Diego, 2006)*, volume 93B, 2025. [arXiv:2411.11208](#).
- [88] Bruce W. Westbury and Paul Zinn-Justin. A uniform trigonometric R -matrix for the exceptional series, 2024. [arXiv:2406.01348](#).
- [87] Paul Zinn-Justin. Integrability and combinatorics. In *Encyclopedia of Mathematical Physics*, volume 3. 2025. chapter of the Encyclopedia of Mathematical Physics. [arXiv:2404.13221](#).
- [86] Yaping Yang and Paul Zinn-Justin. Higher spin representations of the Yangian of $\mathfrak{sl}_2$ and R -matrices, 2024. [arXiv:2403.17433](#).
- [85] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability III: separated descents, 2023. [arXiv:2306.13855](#).
- [84] Paul Zinn-Justin. The *CotangentSchubert* Macaulay2 package. *Journal of Software for Algebra and Geometry*, 14:73–85, 2024. <https://www.unimelb-macaulay2.cloud.edu.au/#tutorial-CotangentSchubert>.
- [83] Alexandr Garbali and Paul Zinn-Justin. Shuffle algebras, lattice paths and the commuting scheme. In *Hypergeometry, Integrability and Lie Theory*. 2022. Editors Erik Koelink, Stefan Kolb, Nicolai Reshetikhin and Bart Vlaar. [arXiv:2110.07155](#).
- [82] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability II: multiplying motivic Segre classes. *Comm. A.M.S.*, 6:68–152, 2026. [arXiv:2102.00563](#).
- [81] Ajeeth Gunna and Paul Zinn-Justin. Vertex models for canonical Grothendieck polynomials and their duals. *Algebraic Combinatorics*, 6(1):109–163, 2023. [arXiv:2009.13172](#).
- [80] Andrew Elvey Price and Paul Zinn-Justin. The six-vertex model on random planar maps revisited. *Journal of Combinatorial Theory, Series A*, 196:105739, 2023. [arXiv:2007.07928](#).

- [79] Paul Zinn-Justin. The trigonometric E_8 R -matrix. *Lett. Math. Phys.*, 110:3279–3305, 2020. [arXiv:2004.02044](#).
- [78] Paul Zinn-Justin. Honeycombs for Hall polynomials. *Electron. J. Combin.*, 27:P2.23, 2020. [arXiv:1909.10720](#).
- [77] Dmitri Bykov and Paul Zinn-Justin. Higher spin $\mathfrak{sl}_2$ R -matrix from equivariant (co)homology. *Lett. Math. Phys.*, 110:2435–2470, 2020. [arXiv:1904.11107](#).
- [76] Mireille Bousquet-Mélou, Andrew Elvey Price, and Paul Zinn-Justin. Eulerian orientations and the six-vertex model on planar maps. In *Proceedings of the 31st Conference on Formal Power Series and Algebraic Combinatorics (Ljubljana, 2019)*, volume 82B, 2020. <http://fpsac2019.fmf.uni-lj.si/resources/Proceedings/129.pdf>, [arXiv:1902.07369](#).
- [75] Iva Halacheva, Allen Knutson, and Paul Zinn-Justin. Restricting Schubert classes to symplectic Grassmannians using self-dual puzzles. In *Proceedings of the 31st Conference on Formal Power Series and Algebraic Combinatorics (Ljubljana, 2019)*, volume 82B, page Art. 83, 2020. <http://fpsac2019.fmf.uni-lj.si/resources/Proceedings/154.pdf>, [arXiv:1811.07581](#).
- [74] Andrew Conway, Anthony Guttman, and Paul Zinn-Justin. 1324-avoiding permutations revisited. *Advances in Applied Mathematics*, 96:312–333, 2018. [arXiv:1709.01248](#).
- [73] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability I: invariant trilinear forms. *Comm. A.M.S.*, 6:1–67, 2026. [arXiv:1706.10019](#).
- [72] Paul Zinn-Justin. Loop models and K -theory. *SIGMA Symmetry Integrability Geom. Methods Appl.*, 14:paper 69, 48, 2018. [arXiv:1612.05361](#).
- [71] Allen Knutson and Paul Zinn-Justin. Grassmann–Grassmann conormal varieties, integrability, and plane partitions. *Annales de l’Institut Fourier*, 69(3):1087–1145, 2019. [arXiv:1612.04465](#).
- [70] Michael Wheeler and Paul Zinn-Justin. Littlewood–Richardson coefficients for Grothendieck polynomials from integrability. *J. Reine Angew. Math.*, 757:159–195, 2019. [arXiv:1607.02396](#).
- [69] Michael Wheeler and Paul Zinn-Justin. Hall polynomials, inverse Kostka polynomials and puzzles. *Journal of Combinatorial Theory, Series A*, 159:107–163, 2018. [arXiv:1603.01815](#).
- [68] Peter Finch, Robert Weston, and Paul Zinn-Justin. Theta function solutions of the quantum Knizhnik–Zamolodchikov–Bernard equation for a face model. *J. Phys. A*, 49(6), 2016. [arXiv:1509.04594](#).
- [67] Michael Wheeler and Paul Zinn-Justin. Refined Cauchy/Littlewood identities and six-vertex model partition functions: III. deformed bosons. *Adv. Math.*, 299:543–600, 2016. [arXiv:1508.02236](#).
- [66] Paul Zinn-Justin. Quiver varieties and the quantum Knizhnik–Zamolodchikov equation. *Theoretical and Mathematical Physics*, 185(3):1741–1758, 2015. [arXiv:1502.01093](#).
- [65] Peter Forrester, L. Dang-Zheng, and Paul Zinn-Justin. Equilibrium problems for Raney densities. *Nonlinearity*, 28(7):2265, 2015. [arXiv:1411.4091](#).
- [64] Anita Ponsaing and Paul Zinn-Justin. Type $\hat{C}$ Brauer loop schemes and loop model with boundaries. *Ann. Henri Poincaré*, 3(2):163–255, 2016. [arXiv:1410.0262](#).
- [63] D. Betea, Michael Wheeler, and Paul Zinn-Justin. Refined Cauchy/Littlewood identities and six-vertex model partition functions: II. proofs and new conjectures. *Journal of Algebraic Combinatorics*, pages 1–49, 2015. [arXiv:1405.7035](#).
- [62] Yacine Ikhlef, Robert Weston, Michael Wheeler, and Paul Zinn-Justin. Discrete holomorphicity and quantized affine algebras. *J. Phys. A*, 46(26):265205, 2013. [arXiv:1302.4649](#).
- [61] Tiago Fonseca and Paul Zinn-Justin. Higher spin polynomial solutions of quantum Knizhnik–Zamolodchikov equation. *Comm. Math. Phys.*, 328(3):1079–1115, 2014. [arXiv:1212.4672](#).
- [60] Roger Behrend, Philippe Di Francesco, and Paul Zinn-Justin. A doubly refined enumeration of alternating sign matrices and descending plane partitions. *J. Combin. Theory Ser. A*, 120(2):409–432, 2013. [arXiv:1202.1520](#).
- [59] Paul Zinn-Justin. Sum rule for the eight-vertex model on its combinatorial line. In *Symmetries, Integrable systems and representations*, volume 40, 2013. Eds Iohara, Morier-Genoud, Rémy. [arXiv:1202.4420](#).

- [58] R. Rimányi, V. Tarasov, A. Varchenko, and Paul Zinn-Justin. Extended Joseph polynomials, quantized conformal blocks, and a q -Selberg type integral. *J. Geom. Phys.*, 62(11):2188–2207, 2012. [arXiv:1110.2187](#).
- [57] Roger Behrend, Philippe Di Francesco, and Paul Zinn-Justin. On the weighted enumeration of alternating sign matrices and descending plane partitions. *J. Combin. Theory Ser. A*, 119(2):331–363, 2012. [arXiv:1103.1176](#).
- [56] Alice Guionnet, Vaughn Jones, Dimitri Shlyakhtenko, and Paul Zinn-Justin. Loop models, random matrices and planar algebras. *Comm. Math. Phys.*, 316(1):45–97, 2012. [arXiv:1012.0619](#).
- [55] Filippo Colomo, Andrei Pronko, and Paul Zinn-Justin. The arctic curve of the domain-wall six-vertex model in its anti-ferroelectric regime. *J. Stat. Mech. Theory Exp.*, page L03002, 2010. [arXiv:1001.2189](#).
- [54] K. Shigechi and Paul Zinn-Justin. Path representation of maximal parabolic Kazhdan–Lusztig polynomials. *J. Pure Appl. Algebra*, 216(11):2533–2548, 2012. [arXiv:1001.1080](#).
- [53] Paul Zinn-Justin. Jucys–Murphy elements and Weingarten matrices. *Lett. Math. Phys.*, 91(2):119–127, 2010. [arXiv:0907.2719](#).
- [52] Paul Zinn-Justin and Jean-Bernard Zuber. Knot theory and matrix integrals. In *The Oxford Handbook of Random Matrix Theory*. 2011. Eds Akemann, Baik and Di Francesco. [arXiv:1006.1812](#).
- [51] Paul Zinn-Justin. A conjectured formula for Fully Packed Loop configurations in a triangle. *Electron. J. Combin.*, 17:Research Paper 107, 2010. [arXiv:0911.4617](#).
- [50] Teo Banica, Benoit Collins, and Paul Zinn-Justin. Spectral analysis of the free orthogonal matrix. *Int. Math. Res. Not.*, 17:3286–3309, 2009. [arXiv:0901.1918](#).
- [49] Tiago Fonseca and Paul Zinn-Justin. On some ground state components of the $O(1)$ loop model. *J. Stat. Mech. Theory Exp.*, page P03025, 2009. [arXiv:0901.1679](#).
- [48] Paul Zinn-Justin. Littlewood–Richardson coefficients and integrable tilings. *Electron. J. Combin.*, 16:Research Paper 12, 2009. [arXiv:0809.2392](#).
- [47] Paul Zinn-Justin. *Six-vertex, loop and tiling models: integrability and combinatorics*. Lambert Academic Publishing, 2009. Habilitation thesis. <http://www.lpthe.jussieu.fr/~pzinn/publi/hdr.pdf>.
- [46] Paul Zinn-Justin. Integrability and combinatorics: selected topics. In *Exact Methods in Low-dimensional Statistical Physics and Quantum Computing*, volume 89. Oxford University Press, 2010. Eds Jacobsen, Ouvry, Pasquier, Serban, and Cugliandolo. <http://www.lpthe.jussieu.fr/~pzinn/semi/intcomb.pdf>.
- [45] Tiago Fonseca and Paul Zinn-Justin. On the doubly refined enumeration of Alternating Sign Matrices and Totally Symmetric Self-Complementary Plane Partitions. *Electron. J. Combin.*, 15:Research Paper 81, 35 pp, 2008. [arXiv:0803.1595](#).
- [44] Jan de Gier, Pavel Pyatov, and Paul Zinn-Justin. Punctured plane partitions and the q -deformed Knizhnik–Zamolodchikov and Hirota equations. *J. Combin. Theory Ser. A*, 116(4):772–794, 2009. [arXiv:0712.3584](#).
- [43] Philippe Di Francesco and Paul Zinn-Justin. Quantum Knizhnik–Zamolodchikov equation: reflecting boundary conditions and combinatorics. *J. Stat. Mech. Theory Exp.*, (12):P12009, 30 pp, 2007. [arXiv:0709.3410](#).
- [42] Alexander Razumov, Yuri Stroganov, and Paul Zinn-Justin. Polynomial solutions of q KZ equation and ground state of XXZ spin chain at $\Delta = -1/2$. *J. Phys. A*, 40(39):11827–11847, 2007. [arXiv:0704.3542](#).
- [41] Philippe Di Francesco and Paul Zinn-Justin. Quantum Knizhnik–Zamolodchikov equation, Totally Symmetric Self-Complementary Plane Partitions and Alternating Sign Matrices. *Theor. Math. Phys.*, 154(3):331–348, 2008. [arXiv:math-ph/0703015](#).
- [40] Paul Zinn-Justin. Loop model with mixed boundary conditions, q KZ equation and alternating sign matrices. *J. Stat. Mech. Theory Exp.*, (1):P01007, 16 pp, 2007. [arXiv:math-ph/0610067](#).
- [39] Allen Knutson and Paul Zinn-Justin. The Brauer loop scheme and orbital varieties. *J. Geom. Phys.*, 78:80–110, 2014. [arXiv:1001.3335](#).
- [38] Paul Zinn-Justin. Proof of the Razumov–Stroganov conjecture for some infinite families of link patterns. *Electron. J. Combin.*, 13(1):Research Paper 110, 15 pp, 2006. [arXiv:math/0607183](#).

- [37] Paul Zinn-Justin. Combinatorial point for fused loop models. *Comm. Math. Phys.*, 272(3):661–682, 2007. [arXiv:math-ph/0603018](#).
- [36] Philippe Di Francesco, Paul Zinn-Justin, and Jean-Bernard Zuber. Sum rules for the ground states of the $O(1)$ loop model on a cylinder and the XXZ spin chain. *J. Stat. Mech.*, page P08011, 2006. [arXiv:math-ph/0603009](#).
- [35] Philippe Di Francesco and Paul Zinn-Justin. From orbital varieties to alternating sign matrices. In *Proceedings of the 18th Conference on Formal Power Series and Algebraic Combinatorics (San Diego, 2006)*, 2006. <http://www-igm.univ-mlv.fr/~fpsac/FPSAC06/SITE06/papers/60.pdf>, [arXiv:math-ph/0512047](#).
- [34] Philippe Di Francesco and Paul Zinn-Justin. Quantum Knizhnik–Zamolodchikov equation, generalized Razumov–Stroganov sum rules and extended Joseph polynomials. *J. Phys. A*, 38(48):L815–L822, 2005. [arXiv:math-ph/0508059](#).
- [33] Allen Knutson and Paul Zinn-Justin. A scheme related to the Brauer loop model. *Adv. Math.*, 214(1):40–77, 2007. [arXiv:math.AG/0503224](#).
- [32] Philippe Di Francesco and Paul Zinn-Justin. Inhomogeneous model of crossing loops and multidegrees of some algebraic varieties. *Comm. Math. Phys.*, 262(2):459–487, 2006. [arXiv:math-ph/0412031](#).
- [31] Philippe Di Francesco and Paul Zinn-Justin. Around the Razumov–Stroganov conjecture: proof of a multi-parameter sum rule. *Electron. J. Combin.*, 12:Research Paper 6, 27 pp, 2005. http://www.combinatorics.org/Volume_12/Abstracts/v12i1r6.html, [arXiv:math-ph/0410061](#).
- [30] Philippe Di Francesco, Paul Zinn-Justin, and Jean-Bernard Zuber. Determinant formulae for some tiling problems and application to fully packed loops. *Ann. Inst. Fourier (Grenoble)*, 55(6):2025–2050, 2005. [arXiv:math-ph/0410002](#).
- [29] Paul Zinn-Justin. Conjectures on the enumeration of alternating links. In *Physical and numerical models in knot theory*, volume 36 of *Ser. Knots Everything*, pages 597–606. World Sci. Publ., Singapore, 2005.
- [28] Jesper Jacobsen and Paul Zinn-Justin. Algebraic Bethe Ansatz for the FPL^2 model. *J. Phys. A*, 37(29):7213–7225, 2004. [arXiv:math-ph/0402008](#).
- [27] Philippe Di Francesco, Paul Zinn-Justin, and Jean-Bernard Zuber. A bijection between classes of fully packed loops and plane partitions. *Electron. J. Combin.*, 11(1):Research Paper 64, 11 pp, 2004. [arXiv:math/0311220](#).
- [26] Paul Zinn-Justin. The asymmetric ABAB matrix model. *Europhys. Lett.*, 64(6):737–742, 2003. [arXiv:hep-th/0308132](#).
- [25] Paul Zinn-Justin and Jean-Bernard Zuber. Matrix integrals and the generation and counting of virtual tangles and links. *J. Knot Theory Ramifications*, 13(3):325–355, 2004. [arXiv:math-ph/0303049](#).
- [24] Gilles Schaeffer and Paul Zinn-Justin. On the asymptotic number of plane curves and alternating knots. *Experiment. Math.*, 13(4):483–493, 2004. [arXiv:math-ph/0304034](#).
- [23] Paul Zinn-Justin and Jean-Bernard Zuber. On some integrals over the $U(N)$ unitary group and their large N limit. *J. Phys. A*, 36(12):3173–3193, 2003. special issue on Random Matrix Theory. [arXiv:math-ph/0209019](#).
- [22] Jesper Jacobsen and Paul Zinn-Justin. Monochromatic path crossing exponents and graph connectivity in 2d percolation. *Phys. Rev. E*, 66(R):055102, 2002. [arXiv:cond-mat/0207063](#).
- [21] Paul Zinn-Justin. HCIZ integral and 2D Toda lattice hierarchy. *Nuclear Phys. B*, 634(3):417–432, 2002. [arXiv:math-ph/0202045](#).
- [20] Paul Zinn-Justin. The influence of boundary conditions in the six-vertex model. unpublished preprint, 2002. [arXiv:cond-mat/0205192](#).
- [19] Jesper Jacobsen and Paul Zinn-Justin. A transfer matrix for the backbone exponent of two-dimensional percolation. *J. Phys. A*, 35(9):2131–2144, 2002. [arXiv:cond-mat/0111374](#).
- [18] Jesper Jacobsen and Paul Zinn-Justin. Matrix models and the enumeration of alternating tangles. *Markov Process. Related Fields*, 9(2):301–310, 2003. proceedings of “Inhomogeneous random systems” (Cergy-Pontoise, 2002).

- [17] Jesper Jacobsen and Paul Zinn-Justin. The combinatorics of alternating tangles: from theory to computerized enumeration. In *Asymptotic combinatorics with application to mathematical physics (St. Petersburg, 2001)*, volume 77 of *NATO Sci. Ser. II Math. Phys. Chem.*, pages 97–112. Kluwer Acad. Publ., Dordrecht, 2002. [arXiv:math-ph/0111011](#).
- [16] Paul Zinn-Justin. The general $O(n)$ quartic matrix model and its application to counting tangles and links. *Comm. Math. Phys.*, 238(1-2):287–304, 2003. [arXiv:math-ph/0106005](#).
- [15] Jesper Jacobsen and Paul Zinn-Justin. A transfer matrix approach to the enumeration of colored links. *J. Knot Theory Ramifications*, 10(8):1233–1267, 2001. [arXiv:math-ph/0104009](#).
- [14] Jesper Jacobsen and Paul Zinn-Justin. A transfer matrix approach to the enumeration of knots. *J. Knot Theory Ramifications*, 11(5):739–758, 2002. [arXiv:math-ph/0102015](#).
- [13] Vladimir Korepin and Paul Zinn-Justin. Inhomogeneous six-vertex model with domain wall boundary conditions and Bethe Ansatz. *J. Math. Phys.*, 43(6):3261–3267, 2002. [arXiv:nlin.SI/0008030](#).
- [12] Paul Zinn-Justin. Six-vertex model with domain wall boundary conditions and one-matrix model. *Phys. Rev. E*, 62(3, part A):3411–3418, 2000. [arXiv:math-ph/0005008](#).
- [11] Vladimir Korepin and Paul Zinn-Justin. Thermodynamic limit of the six-vertex model with domain wall boundary conditions. *J. Phys. A*, 33(40):7053–7066, 2000. [arXiv:cond-mat/0004250](#).
- [10] Paul Zinn-Justin and Jean-Bernard Zuber. On the counting of colored tangles. *J. Knot Theory Ramifications*, 9(8):1127–1141, 2000. [arXiv:math-ph/0002020](#).
- [9] Paul Zinn-Justin. The six-vertex model on random lattices. *Europhys. Lett.*, 50(1):15–21, 2000. [arXiv:cond-mat/9909250](#).
- [8] Paul Zinn-Justin. Some matrix integrals related to knots and links. In *Random matrix models and their applications*, volume 40 of *Math. Sci. Res. Inst. Publ.*, pages 421–438. Cambridge Univ. Press, Cambridge, 2001. [arXiv:math-ph/9910010](#).
- [7] Paul Zinn-Justin and Jean-Bernard Zuber. Matrix integrals and the counting of tangles and links. *Discrete Math.*, 246(1-3):343–360, 2002. [arXiv:math-ph/9904019](#).
- [6] Paul Zinn-Justin. The dilute Potts model on random surfaces. *J. Statist. Phys.*, 98(1-2):j10–264, 2000. [arXiv:cond-mat/9903385](#).
- [5] Paul Zinn-Justin. Adding and multiplying random matrices: a generalization of Voiculescu’s formulas. *Phys. Rev. E* (3), 59(5, part A):4884–4888, 1999. [arXiv:math-ph/9810010](#).
- [4] Vladimir Kazakov and Paul Zinn-Justin. Two-matrix model with ABAB interaction. *Nuclear Phys. B*, 546(3):647–668, 1999. [arXiv:hep-th/9808043](#).
- [3] Paul Zinn-Justin. Nonlinear integral equations for complex affine Toda models associated with simply laced Lie algebras. *J. Phys. A*, 31(31):6747–6770, 1998. [arXiv:hep-th/9712222](#).
- [2] Paul Zinn-Justin and Natan Andrei. The generalized multi-channel Kondo model: thermodynamics and fusion equations. *Nuclear Phys. B*, 528(3):648–682, 1998. [arXiv:cond-mat/9801158](#).
- [1] Paul Zinn-Justin. Universality of correlation functions of Hermitian random matrices in an external field. *Comm. Math. Phys.*, 194(3):631–650, 1998. [arXiv:cond-mat/9705044](#).
- [0] Paul Zinn-Justin. Random Hermitian matrices in an external field. *Nuclear Phys. B*, 497(3):725–732, 1997. [arXiv:cond-mat/9703033](#).